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CivicPulse:
Multi-Agent Intelligent Grievance Redressal & Civic-
Accountability Platform

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Introduction

- CivicPulse is an AI-powered platform that helps citizens report public issues such as potholes, garbage problems, water leaks, and broken streetlights. Users can submit complaints with photos and location details through a mobile phone or computer.
- Municipal grievance systems face a common bottleneck: manual effort to read, categorize, route, and track thousands of complaints , leading to delayed resolutions, misrouted tickets, missed escalations, and eroded public trust.
- Every complaint is automatically classified, deduplicated, prioritized, routed, and tracked to resolution
- Beyond complaint management, CivicPulse serves as an intelligent civic analytics platform, identifying urban problem hotspots, predicting service delays, and supporting smarter city planning through AI-driven insights.

Objectives

- Build a working end-to-end app: citizen submits a complaint (text + photo + GPS) and tracks it through Assigned → In Progress → Resolved.
- Automatic multilingual classification + routing , every complaint reaches the right department without manual sorting.
- Deduplication + severity ranking , merges repeat complaints about the same issue and ranks by real physical severity, not complaint count.
- Predictive SLA-risk model , flags at-risk tickets to staff *before* deadlines are missed.
- Accountability layer , surfaces reopened tickets and recurring complaints at the same location to oversight bodies.
- Policy-insight dashboard , hotspot maps for city planners, corrected for known under-reporting bias in certain areas.
- Deliver a system that's fair, proactive, and transparent , not just another complaint tracker.

Existing System

Several grievance redressal platforms have been developed to help citizens report civic issues and track their status. Most of these systems focus on basic complaint submission, categorization, and status tracking using simple form-based inputs or rule-based routing.

Some large-scale government platforms such as India's CPGRAMS handle complaint intake and routing at scale, but they primarily depend on manual sorting and department-level self-reporting rather than intelligent, automated triage of the complaint's actual content and severity.

Existing research also treats duplicate detection and severity estimation as separate, isolated features, and does not combine them so that repeat complaints are automatically ranked by real-world severity.

However, these systems are generally reactive , they track SLA compliance only after a deadline has already been missed, and mainly stop at logging and routing a ticket rather than predicting risk or verifying real-world resolution.

Most current systems:

- Detect and route complaints, but do not predict SLA-breach risk before it happens.
- Do not merge duplicate complaints with severity-based ranking , count of complaints, not real severity, drives priority.
- Optimize for "was the ticket closed," not "was the problem actually fixed.“
- Do not correct for reporting-equity bias wealthier, more vocal areas get over-represented in resource allocation.
- Track accountability at the department level only, with no independent oversight-facing signal for reopened or recurring tickets.

Research Gap

Deduplication and severity are treated as separate afterthoughts

Existing systems mention duplicate-matching as one feature among many, but none combine it with photo-based severity estimation — so a cluster of duplicate reports is ranked by how many people complained, not by real physical severity (e.g., actual pothole size).

Every reviewed system is reactive, not proactive

Current systems track SLA compliance only after a deadline is already missed. None predict, at the time of intake, the probability that a given ticket will breach its SLA based on the department's own historical resolution-time patterns for that category and location.

No system corrects for reporting-equity bias or tracks real accountability

Reporting is demographically skewed , wealthier, more vocal populations over-report relative to their true issue burden, while poorer, under-represented populations under-report. No existing grievance system corrects hotspot maps for this bias. Separately, no reviewed system tracks reopened-ticket rates or multi-year recurrence at the same location as a signal to independent oversight , only to the department itself, which has no incentive to expose it.

Literature Review

Research Paper	Authors	Methodology	Key Findings	Research Gap
Equity in 311 Reporting: Understanding Socio-Spatial Differentials in the Propensity to Complain (2017)	Constantine E. Kontokosta, Boyeong Hong & Kristi Korsberg	Analyzed 20M+ NYC 311 requests with building-violation, property, and U.S. Census data; used a gradient-boosting regression model to predict expected violations per building, then compared against actual complaint volume.	Reporting is demographically skewed: wealthier, more connected areas over-report relative to true issue burden, while poorer neighborhoods under-report, making raw hotspot maps misleading.	Directly motivates CivicPulse's Gap 3 — the Policy-Insight Agent's census-weighted, equity-corrected hotspot overlay. No AI correction model was proposed in the original paper.
Near-real Time Prediction of Service Requests' SLA Breach (2015)	Sujoy Roychowdhury et al.	Deployed a discrete-time survival-analysis / online ML framework that scores the in-progress probability of a ticket breaching its resolution-time SLA, re-scoring as the ticket ages.	Discriminatory power between requests that will breach vs. resolve on time improves as the ticket progresses; early risk scoring lets staff intervene before the deadline is missed.	Forms the technical basis for CivicPulse's Gap 2 — the SLA-Risk & Escalation Agent — but was built for enterprise IT tickets, not civic grievances, and has no notion of location, category-level equity, or citizen-facing transparency.
AI-driven Vision-based Pothole Detection for Improved Road Safety (2025)	Muhammad Aizat Rasee, Ung Ling Ling, Gloria Jennis Tan, Tan Chi Wee, Ron Buning, Norziana Yahya & Habibah Ismail	Trained a CNN for pothole/normal classification and YOLO for pothole localization and area estimation on a combined 4,681-image road dataset (70/20 train/test split).	The CNN-YOLO pipeline reached 92.85% accuracy and reliably estimated pothole area from a single photo, showing image-based severity scoring is practical at low cost.	Supports CivicPulse's Gap 1 — the Multimodal Severity & Deduplication Agent — but is a single-purpose detector with no duplicate-report clustering or fusion with textual complaint data.

Proposed System

CivicPulse is a **multi-agent architecture**, where each agent owns a distinct responsibility and communicates through shared database state rather than one monolithic model.

SLA-Risk & Escalation Agent : a gradient-boosted survival model predicts breach probability at intake using historical resolution-time distributions, re-scoring daily while a ticket is open.

Escalation & Accountability Agent : rule/statistics-based anomaly detection flags reopened tickets and multi-year recurrence patterns to an independent oversight view, not just the department.

Policy-Insight Agent : aggregates clustered ticket data into hotspot maps with census-weighted equity correction, refreshed via a nightly batch job for city planners.

Backend & Infrastructure:

Backend: FastAPI (Python) / Node+Express, PostgreSQL + PostGIS for geo-data

Vector Search: FAISS or Qdrant (powers deduplication clustering)

Frontend: React (Progressive Web App), Mapbox/Leaflet for hotspot visualization

Notifications: Twilio (SMS), SMTP (email)

DevOps: Docker, GitHub Actions (CI), MLflow (experiment tracking)

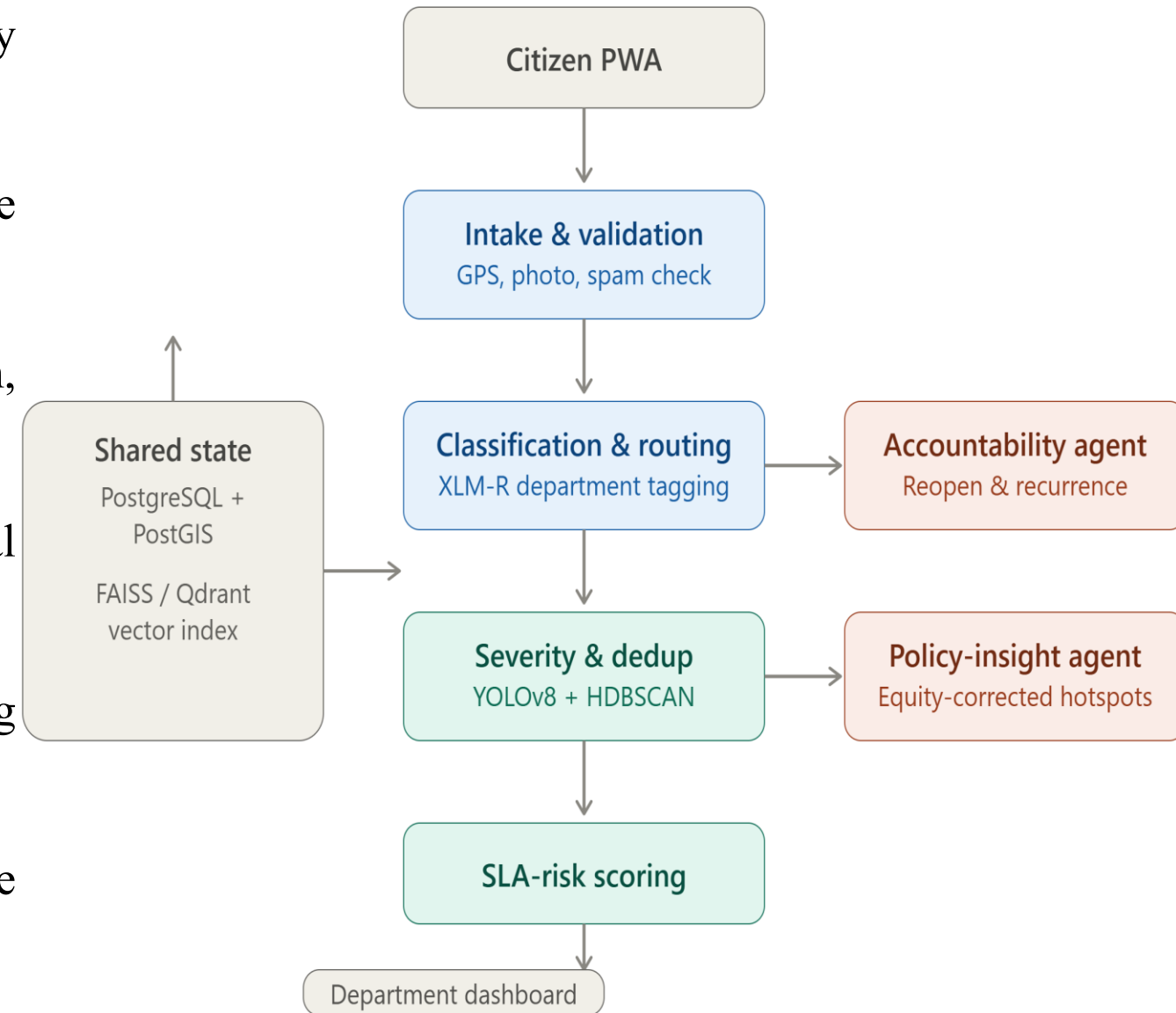
Cloud: AWS/GCP, with an on-prem deployment mode documented

End-to-End Flow:

- Citizen submits complaint (text + photo + GPS) via the PWA.
- Intake Agent validates and rejects/flags spam or out-of-bounds submissions.
- Classification Agent tags department + sub-issue, assigns a queue.
- Severity & Dedup Agent checks for existing similar tickets - merges + re-ranks, or scores severity independently if new.
- SLA-Risk Agent computes and continuously updates a breach-probability score.
- Department staff work the prioritized queue via their dashboard; every status change is logged.
- On closure, citizen gives feedback; if reopened, the Accountability Agent logs it against the department's record.
- Nightly batch job refreshes the Policy-Insight hotspot map with equity correction for city planners.

Expected Outcomes

- Faster, fairer complaint resolution for citizens , no complaint lost or silently ignored.
- Reduced manual triage burden for department staff ,a prioritized queue instead of a raw inbox.
- Fewer missed deadlines : SLA-risk scores flag at-risk tickets before breach, not after.
- Genuinely deduplicated ticket , duplicate reports merged and ranked by real severity, not complaint count.
- Independent, tamper-resistant accountability signal , reopened and recurring complaints surfaced to oversight bodies, not just the department involved.
- Equity-corrected resource allocation , hotspot maps that don't just reward the loudest neighborhoods, giving under-reporting areas fair visibility.
- A transparent audit trail , every classification, merge, and escalation is explainable, building public trust in the system.



1. Citizen PWA

This is the web/mobile application where citizens submit complaints with:

- Complaint description
- Photo
- GPS location

2. Intake & Validation Agent

The first AI agent checks whether:

- GPS location is valid
- Photo is uploaded correctly
- Complaint is not spam
- Required details are available

If everything is correct, the complaint is saved in the database.

3. Classification & Routing Agent

This agent reads the complaint text and determines:

- Complaint category
- Sub-category
- Responsible government department

Example:

- Pothole → Road Department
- Garbage → Sanitation Department
- Water Leakage → Water Department

4. Severity & Duplicate Detection Agent

This agent analyzes the uploaded image.

It:

- Estimates how serious the issue is.
- Checks if the same issue has already been reported nearby.
- Links duplicate complaints instead of creating new ones.

5. SLA Risk Scoring Agent

This agent predicts whether the complaint is likely to be delayed.

If delay is expected:

- Complaint is marked High Priority.
- Officials receive an alert.

6. Accountability Agent

This agent monitors department performance.

It identifies:

- Frequently reopened complaints
- Recurring issues
- Departments missing deadlines

This improves transparency and accountability.

7. Policy Insight Agent

This agent works on historical complaint data.

It:

- Finds complaint hotspots
- Generates city-level reports
- Helps authorities plan maintenance and resource allocation

THANK YOU !